The Matchmakers

Can self-perception predict how often you match?

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Does self-perception predict dating outcomes?

INPUT

How you rate yourself



OUTPUT

How often you match

A speed-dating experiment, 2002–2004

8,378
dates, one per row

123
columns per date

16.5%
of dates ended in a match (1,380)

Source: Fisman & Iyengar, Columbia Business School, via OpenML (SpeedDating)

The features that matter most

Group	Columns	What it captures
Self-ratings (our inputs)	attractive, funny, intelligence, sincere, ambition	How you rate yourself, 1–10
Expectations	expected_num_interested_in_me, expected_num_matches	How many people you expect to like you, and to match with
Partner's view	attractive_o, funny_o, intelligence_o, …	How each date rated you, 1–10 (a comparison, not an input)
Outcome	match	1 if both said yes, 0 otherwise

First rows: one person, three dates

SELF-RATINGS · INPUTS							PARTNER'S VIEW	
Date	**Attr.**	**Fun**	**Intel.**	**Sinc.**	**Amb.**	**Exp. matches**	**Attr. by date**	**Match**
1	6	8	8	8	7	4	6	0 (no)
2	6	8	8	8	7	4	7	0 (no)
3	6	8	8	8	7	4	10	1 (yes)

Selected columns from the dataset's first three rows

Limitations we'll have to handle

Imbalanced outcome

83.5% of dates don't match, so always guessing "no" is 83.5% accurate. We'll judge models on ROC-AUC and F1.

Missing values

87.5% of rows have at least one blank field. We'll impute or drop per feature and track the effect.

Repeated people

Each person spans many rows with no ID column. We'll split train and test by person so nobody is memorized.

Narrow sample

Columbia grad students, 2002–2004, opposite-sex pairings, all self-reported. Results may not carry over to today's apps.

Our planned approach

01
Select
The five self-ratings as inputs; match as the label

02
Clean
Handle missing values, check ranges, scale features

03
Model
Logistic regression as an interpretable baseline, then a decision tree and a random forest

04
Evaluate
Cross-validation split by person; ROC-AUC, F1, precision and recall

How we build on prior research

FISMAN ET AL., 2006

Same data, different question

Studied what people value in a partner: women weighted intelligence and race more; men responded more to physical attractiveness.

JOEL, EASTWICK & FINKEL, 2017

Machine learning on self-reports

Random forests on 100+ self-reported traits predicted how desired a person was overall, but not who would click with whom.

OUR PROJECT

Self-view only

Just five self-ratings, aimed at a person-level outcome: the part Joel et al. found predictable. Then we compare against how dates rated each person.

Fisman et al., Quarterly Journal of Economics (2006) · Joel et al., Psychological Science (2017)

Stretch goals, if time allows

1 **Stepwise feature selection**
Search all 123 columns for the features most predictive of a match.

2 **The perception gap**
Train the same models on how dates rated each person, and measure how far self-ratings sit from others' ratings.

3 **Split by gender**
Test whether self-perception predicts matches differently for men and women, echoing Fisman et al.